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Task 1: Install Apache Kafka on our Local Machine.

Download Kafka: The official Apache Kafka download page and download the latest binary release.

Extract Files: Unzip the downloaded file to a clean folder on your drive (e.g., C:\kafka on Windows or /usr/local/kafka on Mac/Linux).

```
C:\kafka\bin\windows>kafka-topics.bat --version
3.7.1
C:\kafka\bin\windows>|
```

Generate a Cluster ID (KRaft Mode): Open your terminal/command prompt, navigate to your Kafka folder, and run:

Step 2: Start Zookeeper and Kafka Server

Command : cd C:\kafka

```
.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
```

[illegible]

Step 2 : Start Kafka Broker

Command : cd C:\kafka

.\bin\windows\kafka-server-start.bat .\config\server.properties

```
PS C:\Users\Gagan Namdev> cd C:\kafka
PS C:\kafka> .\bin\windows\kafka-server-start.bat .\config\server.properties
[2026-08-20 11:56:14,410] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2026-08-20 11:56:14,673] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiation (org.apache.zookeeper.common.X509Util)
[2026-08-20 11:56:14,734] INFO starting (kafka.server.KafkaServer)
[2026-08-20 11:56:14,734] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2026-08-20 11:56:14,746] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181. (kafka.zookeeper.ZooKeeperClient)
[2026-08-20 11:56:14,754] INFO Client environment:zookeeper.version=3.8.4-9316c2a7a97e1666d8f4593f34dd6fc36ecc436c, built on 2024-02-12 22:16 UTC (org.apache.zookeeper.ZooKeeper)
[2026-08-20 11:56:14,754] INFO Client environment:host.name=DESKTOP-FBTABVO (org.apache.zookeeper.ZooKeeper)
[2026-08-20 11:56:14,754] INFO Client environment:java.version=11.0.30 (org.apache.zookeeper.ZooKeeper)
[2026-08-20 11:56:14,754] INFO Client environment:java.vendor=Oracle Corporation (org.apache.zookeeper.ZooKeeper)
[2026-08-20 11:56:14,754] INFO Client environment:java.home=C:\Program Files\Java\jdk-11.0.30 (org.apache.zookeeper.ZooKeeper)
[2026-08-20 11:56:14,754] INFO Client environment:java.class.path=C:\kafka\libs\activation-1.1.1.jar;C:\kafka\libs\aoopalliance-repackaged-2.6.1.jar;C:\kafka\libs\argparse4j-0.7.0.jar;C:\kafka\libs\audience-annotations-0.12.0.jar;C:\kafka\libs\caffeine-2.9.3.jar;C:\kafka\libs\checker-qual-3.19.0.jar;C:\kafka\libs\commons-beanutils-1.9.4.jar;C:\kafka\libs\commons-cli-1.4.jar;C:\kafka\libs\commons-collections-3.2.2.jar;C:\kafka\libs\commons-digester-2.1.jar;C:\kafka\libs\commons-io-2.11.0.jar;C:\kafka\libs\commons-lang3-3.8.1.jar;C:\kafka\libs\commons-logging-1.2.jar;C:\kafka\libs\commons-validator-1.7.jar;C:\kafka\libs\connect-api-3.7.1.jar;C:\kafka\libs\connect-basic-auth-extension-3.7.1.jar;C:\kafka\libs\connect-file-3.7.1.jar;C:\kafka\libs\connect-json-3.7.1.jar;C:\kafka\libs\connect-mirror-3.7.1.jar;C:\kafka\libs\connect-mirror-client-3.7.1.jar;C:\kafka\libs\connect-runtime-3.7.1.jar;C:\kafka\libs\connect-transforms-3.7.1.jar;C:\kafka\libs\error-prone-annotations-2.10.0.jar;C:\kafka\libs\hk2-api-2.6.1.jar;C:\kafka\libs\hk2-locator-2.6.1.jar;C:\kafka\libs\hk2-utils-2.6.1.jar;C:\kafka\libs\jackson-annotations-2.16.2.jar;C:\kafka\libs\jackson-core-2.16.2.jar;C:\kafka\libs\jackson-databind-2.16.2.jar;C:\kafka\libs\jackson-dataformat-csv-2.16.2.jar;C:\kafka\libs\jackson-datatype-jdk8-2.16.2.jar;C:\kafka\libs\jackson-jaxrs-base-2.16.2.jar;C:\kafka\libs\jackson-jaxrs-json-provider-2.16.2.jar;C:\kafka\libs\jackson-module-jaxb-annotations-2.16.2.jar;C:\kafka\libs\jackson-module-scala-2.12-2.16.2.jar;C:\kafka\libs\jakarta.activation-api-1.2.2.jar;C:\kafka\libs\jakarta.annotation-api-1.3.5.jar;C:\kafka\libs\jakarta.inject-2.6.1.jar;C:\kafka\libs\jakarta.validation-api-2.0.2.jar;C:\kafka\libs\jakarta.ws.rs-api-2.1.6.jar;C:\kafka\libs\jakarta.xml.bind-api-2.3.3.jar;C:\kafka\libs\javassist-3.29.2-GA.jar;C:\kafka\libs\javax.activation-api-1.2.0.jar;C:\kafka\libs\javax.annotation-api-1.3.2.jar;C:\kafka\libs\javax.servlet-api-3.1.0.jar;C:\kafka\libs\javax.ws.rs-api-2.1.1.jar;C:\kafka\libs\jaxb-api-2.3.1.jar;C:\kafka\libs\jersey-client-2.39.1.jar;C:\kafka\libs\jersey-common-2.39.1.jar;C:\kafka\libs\jersey-container-servlet-2.39.1.jar;C:\kafka\libs\jersey-container-servlet-core-2.39.1.jar;C:\kafka\libs\jersey-hk2-2.39.1.jar;C:\kafka\libs\jersey-server-2.39.1.jar;C:\kafka\libs\jetty-client-9.4.54.v20240208.jar;C:\kafka\libs\jetty-continuation-9.4.54.v20240208.jar;C:\kafka\libs\jetty-http-9.4.54.v20240208.jar;C:\kafka\libs\jetty-io-9.4.54.v20240208.jar;C:\kafka\libs\jetty-security-9.4.54.v20240208.jar;C:\kafka\libs\jetty-server-9.4.54.v20240208.jar;C:\kafka\libs\jetty-servlet-9.4.54.v20240208.jar;C:\kafka\libs\jetty-servlet-security-9.4.54.v20240208.jar;C:\kafka\libs\jetty-util-9.4.54.v20240208.jar;C:\kafka\libs\jetty-util-ajax-9.4.54.v20240208.jar;C:\kafka\libs\jline-3.25.1.jar;C:\kafka\libs\jopt-simple-5.0.4.jar;C:\kafka\libs\jose4j-0.9.4.jar;C:\kafka\libs\jsr305-3.0.2.jar;C:\kafka\libs\kafka-clients-3.7.1.jar;C:\kafka\libs\kafka-group
```

Task 2 : Create a topic named student-events.

Command : .\bin\windows\kafka-topics.bat --create --topic student-events

--bootstrap-server localhost:9092 --partitions 1 --replication-factor 1

```
PS C:\Users\Gagan Namdev> cd C:\kafka
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic student-events --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic student-events.
PS C:\kafka> |
```

Verify the topic is created by listing it:

Command : .\bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092

```
PS C:\kafka> .\bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
student-events
PS C:\kafka> |
```

Task 3 : Create a producer and publish 5 messages to the topic.

Command : .\bin\windows\kafka-console-producer.bat --topic student-events

--bootstrap-server localhost:9092

```
PS C:\kafka> .\bin\windows\kafka-console-producer.bat --topic student-events --bootstrap-server localhost:9092
>Message 1: Student Alice registered
>Message 2: Student Ankit submitted assignment
>Message 3: Student somya cleared exam
>Message 4: Student David updated profile
>Message 5: Student Emma logged out
```

Task 4 : Start a consumer and verify all the 5 messages are received.

Command : `.\bin\windows\kafka-console-consumer.bat --topic student-events --from-beginning --bootstrap-server localhost:9092`

```
PS C:\kafka> .\bin\windows\kafka-console-consumer.bat --topic student-events --from-beginning --bootstrap-server localhost:9092
```

```
Message 1: Student Alice registered
Message 2: Student Ankit submitted assignment
Message 3: Student somya cleared exam
Message 4: Student David updated profile
Message 5: Student Emma logged out
```

Task 5 : Create another topic with 3 partitions and observe the partition details.

Command : `.\bin\windows\kafka-topics.bat --create --topic multi-partition-topic --bootstrap-server localhost:9092 --partitions 3 --replication-factor 1`

```
PS C:\Users\Gagan Namdev> cd C:\kafka
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic multi-partition-topic --bootstrap-server localhost:9092 --partitions 3 --replication-factor 1
Created topic multi-partition-topic.
PS C:\kafka> |
```

Now, describe the topic to observe how Kafka structures the partition details:

Command : `.\bin\windows\kafka-topics.bat --describe --topic multi-partition-topic --bootstrap-server localhost:9092`

```
PS C:\kafka> .\bin\windows\kafka-topics.bat --describe --topic multi-partition-topic --bootstrap-server localhost:9092
Topic: multi-partition-topic TopicId: nVTM9yFFThWldr1lg7wE_g PartitionCount: 3 ReplicationFactor: 1 Configs:
Topic: multi-partition-topic Partition: 0 Leader: 0 Replicas: 0 Isr: 0
Topic: multi-partition-topic Partition: 1 Leader: 0 Replicas: 0 Isr: 0
Topic: multi-partition-topic Partition: 2 Leader: 0 Replicas: 0 Isr: 0
PS C:\kafka> |
```

Task 6 : Also, create two consumers in the same consumer group and observe how Kafka distributes messages across them.

Step 1 : Terminal A (Consumer 1) and join a consumer group named student-group:

Command : `kafka-console-consumer.bat --topic assignment-topic --bootstrap-server localhost:9092 --group student-group`

```
C:\kafka\bin\windows>kafka-console-consumer.bat --topic assignment-topic --bootstrap-server localhost:9092 --group student-group
Hello to Partition B
Hello to Partition C
Message for User D
Message for User E
Message for User F
```

Step 2 : Terminal B (Consumer 2) and join the *same* consumer group:

Command : `kafka-console-consumer.bat --topic assignment-topic --bootstrap-server localhost:9092 --group student-group`

```
C:\kafka\bin\windows>kafka-console-consumer.bat --topic assignment-topic --bootstrap-server localhost:9092 --group student-group
Hello to Partition A
|
```

Step 3 : Terminal C (Producer) and send 6-7 random messages to this topic:

Command : `kafka-console-producer.bat --topic assignment-topic --bootstrap-server localhost:9092 --property parse.key=true --property key.separator=:`

```
C:\kafka\bin\windows>kafka-console-producer.bat --topic assignment-topic --bootstrap-server localhost:9092 --property parse.key=true --property key.separator=:
>k1:Hello to Partition A
>k2:Hello to Partition B
>k3:Hello to Partition C
>k4:Message for User D
>k5:Message for User E
>k6:Message for User F
>
```
